Viscometer



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Viscometer

What it is used for

A viscometer measures the viscosity of liquids, indicating their resistance to flow. Select the appropriate instrument and method for the sample to obtain consistent and meaningful results.

Main parts

- Viscometer main unit and display
- Spindle, rotor, capillary, or measuring probe as applicable
- Sample container or measurement tube
- Temperature sensor or temperature-controlled bath, if fitted
- Speed control and measurement buttons
- Stand, clamp, or sample holder
- Calibration fluid and cleaning accessories

Before you start

1. Place the instrument on a stable, level surface away from vibration.
2. Inspect the instrument, spindle, probe, and cables for damage.
3. Select the correct spindle or measuring system for the sample and expected viscosity.
4. Prepare a clean sample container with sufficient sample volume.
5. Bring the sample to the specified test temperature and allow it to stabilize.
6. Check calibration status and verify zero setting where applicable.
7. Confirm that the sample is mixed uniformly without introducing air bubbles.

How to use

1. Switch on the viscometer and select the required measurement mode.
2. Enter the test temperature, spindle type, speed, or method settings as applicable.
3. Attach the clean spindle or probe securely before starting the measurement.
4. Place the sample container in the correct position and set the required immersion depth.
5. Start the instrument at the recommended speed and allow the reading to stabilize.
6. Check that the displayed torque or measurement indicator is within the instrument's valid range.
7. Record viscosity, units, temperature, spindle or probe, and operating speed.
8. Repeat the measurement if required using the same controlled test conditions.
9. Stop the instrument and remove the spindle or probe carefully.
10. Clean and dry the measuring parts according to the manufacturer's instructions.

⚠ Safety

- Wear suitable gloves and eye protection when handling chemical or hot samples.
- Keep liquids away from electrical controls and connectors.
- Do not touch or adjust a moving spindle during operation.
- Avoid contact with hot samples, baths, or heated components.
- Use only approved cleaning agents compatible with the instrument and sample.

Care & storage

- Clean the spindle, probe, and sample holder immediately after use.
- Handle measuring spindles carefully to prevent bending or surface damage.
- Store the instrument in a dry, dust-free location with accessories protected.
- Verify calibration using suitable standards at the recommended intervals.

Common mistakes to avoid

- Measuring at an incorrect or unstable sample temperature.
- Selecting the wrong spindle, speed, or measurement range.
- Introducing air bubbles or failing to mix the sample uniformly.
- Recording readings before they stabilize or outside the valid torque range.
- Leaving residue on the spindle and affecting subsequent measurements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.